

Chronic Kidney Disease Screening in Patients With Coronary Heart Disease

The Multinational INTERASPIRE Study

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ABSTRACT

BACKGROUND Chronic kidney disease (CKD) is an important risk factor for the progression of coronary artery disease (CAD).

OBJECTIVES The purposes of this study were to quantify the prevalence of CKD in CAD patients from 14 countries from all World Health Organization regions and to evaluate the prognostic value of estimated glomerular filtration rate (eGFR) and urinary albumin/creatinine ratio (UACR).

METHODS A total of 4,548 patients with CAD were included (79.6% were males; age range: 18-80 years). They were assessed for eGFR and UACR 6 to 24 months after the CAD diagnosis. Complete information on kidney function and cardio-renal protective therapy was available for 3,865 patients and follow-up data after a median of 1 year were available for 3,577 (92.5%).

RESULTS CKD according to the Kidney Disease Improving Global Outcomes classification was present in 32% of whom 19.7% were classified as low-moderate, 6.9% as high, and 5.6% as very high risk. Without UACR, 51.3% of them would have been undetected. The primary event, first of cardiovascular death, myocardial infarction, stroke, and hospitalization for heart failure, was observed in 7.9%, with the highest incidence in the Kidney Disease Improving Global Outcomes high-risk group (men: 13.0%; women: 11.8%). This relationship was independent of other risk factors and evident soon after the index examination. Only a minority of the patients received adequate cardio-renal protective therapy.

CONCLUSIONS Early screening for CKD in patients with CAD is important and should preferably include both eGFR and UACR to provide a complete diagnosis. Without UACR, half of those with CKD would remain undetected. Treatment with cardio-renal protective therapy was low, providing great potential for improvement. (JACC. 2026;■:■-■) © 2026 by the American College of Cardiology Foundation.

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ABBREVIATIONS
AND ACRONYMS

CAD = coronary artery disease

CKD = chronic kidney disease

eGFR = estimated glomerular
filtration rateHbA1c = glycated hemoglobin
A1cKDIGO = Kidney Disease:
Improving Global OutcomesRAAS = Renin Angiotensin
Aldosterone SystemUACR = urinary albumin
creatinine ratioWHO = World Health
Organization

Globally cardiovascular disease remains a leading cause of mortality, principally due to coronary artery disease (CAD).^{1,2} Chronic kidney disease (CKD) is an independent risk factor for the development of CAD and results in a poorer prognosis.³ According to a recent analysis, the global number of deaths from cardiovascular disease attributable to CKD increased significantly between 1990 and 2021, a trend especially pronounced in the elderly and in regions with a middle socio-demographic index.⁴

CKD is defined by the Kidney Disease: Improving Global Outcomes (KDIGO) as a structural or functional kidney abnormality persisting for more than 3 months with implications for health.⁵ CKD increases with age from 12% in patients in their forties to 34% in patients in their seventies.^{6,7} Both the incidence and prevalence of CKD are rising, particularly in low- and middle-income countries.⁸⁻¹⁰ The frequent combination of CAD and CKD has been highlighted in a state-of-the-art review by Sarnak et al.¹¹ This review, however, primarily elaborated on when and how to screen for CAD in patients with CKD. Because CKD, usually asymptomatic, is increasing and associated with poorer prognosis in patients with CAD,³ screening for renal

involvement in such patients is essential given the availability of new treatment modalities.

Optimal screening for CKD is a combination of the estimated glomerular filtration rate (eGFR) and the urinary albuminuria creatinine ratio (UACR), an important marker of kidney and vascular damage. In the U.S. National Health and Nutrition Examination Surveys, comprising adults with hypertension or diabetes, approximately two-thirds of patients with albuminuria were undetected due to lack of testing.¹² Albuminuria diagnosis supports the identification of patients at risk of cardiovascular disease and CKD and presents an early opportunity to intervene with cardio-renal protective therapy to improve patient outcomes.¹³ Yet, many international guidelines for the management of CAD do not emphasize the importance of routinely evaluating for moderate and severe albuminuria as a routine screening tool for CKD.¹⁴ A European survey based on the EUROASPIRE (European Action on Secondary Prevention through Intervention to Reduce Events) study found that without adding UACR to eGFR almost one-half of CAD patients with CKD would have remained undiagnosed.¹⁵ Given that the increase in CKD-related mortality is particularly apparent in middle-income regions,⁴ it is important to promote CKD screening not only in European countries but specifically to CAD patients in these countries as well.

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The authors attest they are in compliance with human studies committees and animal welfare regulations of the authors' institutions and Food and Drug Administration guidelines, including patient consent where appropriate. For more information, visit the [Author Center](#).

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The INTERASPIRE (International Action on Secondary Prevention through Intervention to Reduce Events) study, a survey of secondary prevention in patients with CAD, was carried out in collaboration with the World Heart Federation and regional and national cardiology societies according to methodology established in the EUROASPIRE study.¹⁶ The objectives were to: 1) study the practice of screening for CKD in patients with CAD in 6 World Health Organization (WHO) regions; 2) determine the prognostic implication of CKD; and 3) investigate the use of cardio-renal protective therapy in CAD patients with CKD.

MATERIALS AND METHODS

STUDY DESIGN AND PARTICIPANTS. INTERASPIRE, a longitudinal, cross-sectional study of adults with CAD was conducted from 2020-2023, in 14 countries from all 6 WHO regions: Africa (Kenya, Nigeria, and Tanzania), Americas (Argentina and Colombia), Eastern Mediterranean Region (Egypt and United Arab Emirates), Europe (Poland and Portugal), South East Asia (Indonesia), and Western Pacific Region (China, Malaysia, Philippines, and Singapore). Ethics approvals were obtained in all hospital centers and all patients provided written informed consent according to local practice.

A detailed description of INTERASPIRE has been presented elsewhere.^{16,17} In brief, 4,548 (21% were women) consecutive patients, 18 to 80 years of age, hospitalized with a first or recurrent CAD event (elective or emergency coronary artery bypass graft, elective or emergency percutaneous coronary intervention, acute myocardial infarction, or unstable angina/acute myocardial ischemia), were identified. They were invited to a study visit between 6 months and 2 years after the index event (except for Africa where the timeframe was extended to 36 months to enable recruitment).

DATA COLLECTION. Data were collected by centrally trained research staff using standardized methods and instruments. Information on their case history, demographics, health behaviors, risk factor management, and all prescribed medication was obtained from both their medical records during the index CAD hospitalization and from the subsequent patient interviews conducted during the INTERASPIRE study visit. The patients brought their prescribed medications to the visit.

The following standardized assessments were performed at the time of the study visit. Smoking was defined as self-reported smoking, and/or a breath carbon monoxide exceeding 10 ppm by means of a

Smokerlyzer (Bedfont Scientific, Model Micro +). Persistent smoking was defined as smoking at the time of the interview among those who smoked in the month before the index event.

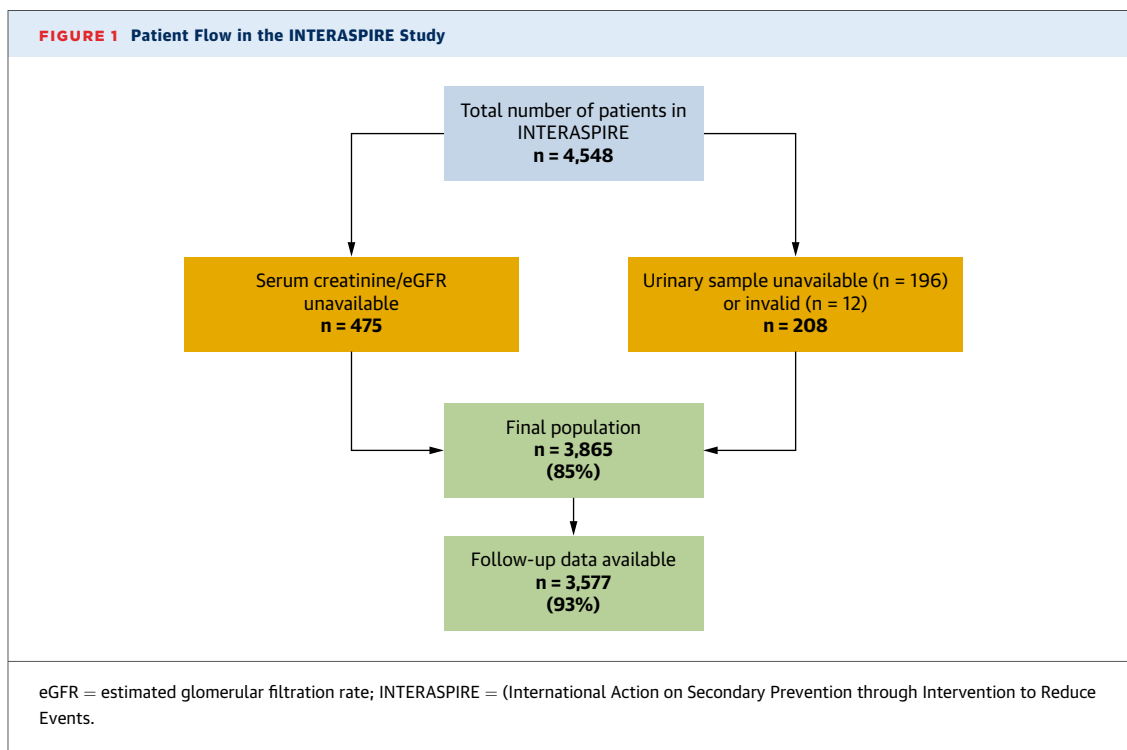
Height and weight were measured in light indoor clothes without shoes using a calibrated measuring stick and scale. Obesity was defined as a body mass index ≥ 30 kg/m².

Waist circumference was measured with the patient standing using a metal tape applied at the point between the lowest rim of the rib cage and the tip of the superior iliac crest. A waist-to-height ratio was considered high if ≥ 0.6 .

Blood pressure was measured twice on the right upper arm after 5 minutes in a sitting position using a digital sphygmomanometer (Omron Health Care) and the mean of both measurements was used for the analysis.

Venous (fasting) blood was drawn in a sitting position for serum total and high-density lipoprotein cholesterol, triglycerides, and creatinine. Fasting glucose was measured in plasma, and glycated hemoglobin (HbA1c) in whole blood. Samples were taken with light stasis into a tube containing clot activator (Vacutainer SST II Advanced, Becton Dickinson) for lipid assays and creatinine, fluoride citrate sampling tube for plasma glucose (fasting and 2-hour glucose), and a potassium EDTA tube (Vacutainer K2EDTA) for the HbA1c assay. After centrifugation, serum and plasma were aliquoted and stored together with EDTA tubes with whole blood at a minimum of -70°C . Low-density lipoprotein cholesterol was calculated using the Friedewald formula if the triglyceride level was <4.5 mmol/L (<400 mg/dL). An oral glucose tolerance test (75 g glucose dissolved in 200 mL water) was performed in patients free from known diabetes with fasting and 2-hour plasma glucose measured in the local laboratories.

The laboratory analyses for all countries, except 4, were conducted at a Central Laboratory (Department of Government Services, Finnish Institute for Health and Welfare [THL]), accredited by the Finnish Accreditation Service and in compliance with the requirements of the standard SFS-EN ISO/IEC 17025:2005. Frozen blood samples were transported on dry ice from the national laboratories to the Central Laboratory. All measurements were performed on a clinical chemistry analyzer (Architect c8000; Abbott Laboratories). The following methods were used: enzymatic methods for total and high-density lipoprotein cholesterol, creatinine, triglycerides, and HbA1c, and an enzymatic hexokinase method for glucose. Creatinine and albumin in urine were analyzed locally.



The laboratory analyses of the samples from China, Egypt, Indonesia, and United Arab Emirates were performed by an accredited laboratory in each

country. Standardization of results between these national laboratories and the Central Laboratory in Helsinki was performed.

TABLE 1 Pertinent Clinical Characteristics of the Patient Population

	Total Population (N = 3,865)	Men (n = 3,075)	Women (n = 790)
Age, y	60.2 ± 10.2	59.6 ± 10.2	62.5 ± 10.0
Index event/history			
Myocardial infarction	59.2% (2,290/3,865)	61.0% (1,877/3,075)	52.3% (413/790)
Percutaneous coronary intervention	37.6% (1,452/3,865)	37.9% (1,165/3,075)	36.3% (287/790)
Coronary bypass surgery	12.0% (463/3,865)	13.0% (400/3,075)	8.0% (63/790)
History of stroke	3.1% (119/3,865)	3.0% (91/3,075)	3.5% (28/790)
History of diabetes	43.7% (1,688/3,865)	42.9% (1,318/3,075)	46.8% (370/790)
Risk factors			
Smoking			
Current	16.3% (631/3,865)	19.3% (592/3,075)	4.9% (39/790)
Persistent	50.8% (580/1,142)	51.2% (544/1,062)	45.0% (36/80)
Body mass index, kg/m ²	27.2 (4.6)	27.0 (4.4)	27.8 (5.5)
Obesity	23.8% (918/3,858)	21.8% (669/3,071)	31.6% (249/787)
Waist-to-height ratio	0.57 (0.08)	0.57 (0.07)	0.59 (0.09)
Sufficient physical activity ^a	37.7% (1,457/3,865)	40.6% (1,247/3,075)	26.6% (210/790)
Waist circumference, cm	95.4 (12.8)	96.0 (12.4)	92.9 (14.1)
Newly detected diabetes	12.0% (229/1,902)	11.7% (179/1,535)	13.6% (50/367)
Newly detected IGT	27.3% (520/1,902)	26.4% (405/1,535)	31.3% (115/367)

Values are mean ± SD or proportion (n/N). ^aPerforming regular physical activity ≥30 min on average 5 times a week.
IGT = impaired glucose tolerance.

DEFINITIONS. CKD categories were, according to KDIGO, defined by means of moderate or severe albuminuria (UACR: 30-299 mg/g or ≥300 mg/g) and/or eGFR <60 mL/min/1.73 m². eGFR was calculated using the Chronic Kidney Disease Epidemiology Collaboration 2021 formula.¹⁸ eGFR and UACR measurements were used to define KDIGO CKD categories.⁵

Prescribed pharmacologic therapies were based on self-reported use at the interview. Previously known diabetes was defined as a self-reported history of type 2 diabetes or use of any glucose-lowering medication. Among those without previous diabetes, newly detected dysglycemia (impaired glucose tolerance or type 2 diabetes) was based on an oral glucose tolerance test and defined according to WHO.¹⁹

OUTCOMES. Follow-up of the patients was performed 1 year after the INTERASPIRE study visit. Information on fatal and nonfatal cardiovascular events was primarily collected from the patients' medical records, registries, or databases or by contacting the patients' family doctor. Information sought included vital status, date and cause of death (coronary heart disease, stroke, other vascular, cancer, or other causes), and new hospital admissions

FIGURE 2 Distribution on KDIGO According to the Combined Use of eGFR and UACR

eGFR (ml/min/1.73 m ³)	UACR			Total	
	<30 mg/g	30-299 mg/g	≥300 mg/g		
≥90	N = 1,452 37.6%	N = 277 7.2%	N = 47 1.2%	N = 1,776	
60-89	N = 1,169 30.2%	N = 263 6.8%	N = 51 1.3%	N = 1,483	
45-59	N = 221 5.7%	N = 95 2.5%	N = 35 0.9%	N = 351	
30-44	N = 72 1.9%	N = 50 1.3%	N = 31 0.8%	N = 153	
15-29	N = 22 0.6%	N = 13 0.3%	N = 23 0.6%	N = 58	
<15	N = 13 0.3%	N = 5 0.1%	N = 26 0.7%	N = 44	
Total	N = 2,949	N = 703	N = 213	N = 3,865	

In each box the number of patients and the proportion is presented. The total proportions in each risk group are summarized to the right. KDIGO risk categories: green = low; Yellow = moderately increased; Orange = high; Red = very high. eGFR = estimated glomerular filtration rate; KDIGO = Kidney Disease: Improving Global Outcomes; UACR = urinary albumin creatinine ratio.

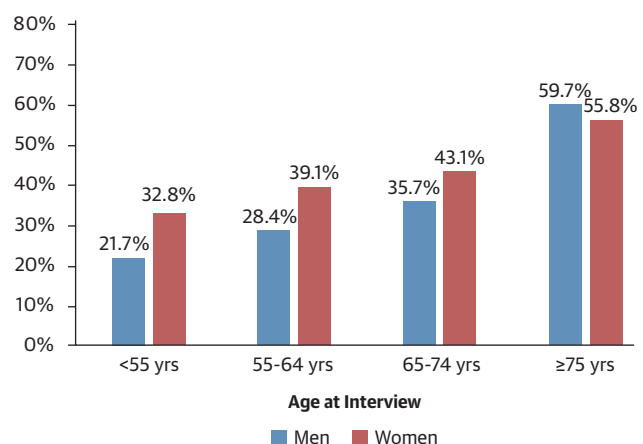
after the baseline interview. The information was recorded using a standardized online questionnaire.

A composite of the first of cardiovascular death, hospitalization for nonfatal myocardial infarction, stroke, or hospitalization for heart failure served as the primary outcome. A secondary endpoint, including all variables in the primary endpoint and, in addition, coronary bypass surgery and percutaneous coronary interventions, was also analyzed.

DATA MANAGEMENT. Data were collected electronically and submitted online to the data management center in the EURObservational Research Programme, European Heart House, Sophia Antipolis, France, for Argentina and Malaysia, and for the other countries, ARO Specialist Data Centre Services (previously AIMES), Liverpool, United Kingdom. Data were checked for completeness, internal consistency, and accuracy. All data were stored under the provisions of the national data protection regulations.

STATISTICAL ANALYSES. Common descriptive statistics (proportion, mean $\pm$ SD, median [IQR]) were used to describe the distributions of categorical and quantitative variables. The relationship between CKD stage, characterized according to the KDIGO classification system, and the risk of the primary and secondary outcomes was investigated by constructing Kaplan-Meier curves and by fitting Cox proportional hazards regression models, adjusted for age at interview, gender, history of myocardial infarction,

history of percutaneous coronary intervention, history of coronary bypass surgery, history of stroke, history of diabetes, smoking, waist-to-height ratio, self-reported physical activity, systolic blood pressure, fasting glucose, HbA1c, low-density lipoprotein cholesterol, and the use of aspirin, beta-blockers, Renin Angiotensin Aldosterone System (RAAS) blockers, or statins. Event-free survival time in these Cox models was calculated as the time between the

FIGURE 3 Distribution of the Prevalence of CKD According to KDIGO by Age and Gender

CKD = chronic kidney disease; other abbreviation as in Figure 2.

TABLE 2 Distribution of Patient Characteristics at Baseline by KDIGO Risk Category

	KDIGO Risk Category		
	Low Risk (n = 2,621)	Moderately Increased Risk (n = 761)	High or Very High Risk (n = 483)
Female	17.9% (468/2,621)	23.0% (175/761)	30.4% (147/483)
Age, y	58.9 ± 10.0	62.1 ± 10.4	64.2 ± 9.5
Ethnicity			
African	8.2% (215/2,621)	14.3% (109/761)	18.0% (87/483)
Asian	54.8% (1,437/2,621)	55.7 (424/761)	52.4% (253/483)
European	22.4% (586/2,621)	20.0% (152/761)	20.1% (97/483)
Latino/Hispanic	12.1% (317/2,621)	8.5% (65/761)	6.4% (31/483)
Other	2.5% (66/2,621)	1.4% (11/761)	3.1 (15/483)
Index event/history			
Myocardial infarction	59.0% (1,546/2,621)	59.0% (449/761)	61.1% (295/483)
Percutaneous coronary intervention	37.6% (986/2,621)	37.6% (286/761)	37.3% (180/483)
Coronary bypass surgery	11.6% (305/2,621)	12.2% (93/761)	13.5% (65/483)
History of stroke	2.1% (55/2,621)	4.2% (32/761)	6.6% (32/483)
History of diabetes	36.6% (958/2,621)	53.7% (409/761)	66.5% (321/483)
Risk factors			
Current smoking	17.7% (463/2,621)	15.6% (119/761)	10.1% (49/483)
Persistent smoking	50.1% (427/852)	55.6% (114/205)	45.9% (39/85)
Body mass index ≥30 kg/m ²	22.8% (598/2,618)	24.6% (187/761)	27.8% (133/479)
Waist-to-height ratio ≥0.6	30.8% (803/2,603)	38.8% (293/756)	44.4% (212/478)
Sufficient physical activity	40.1% (1,050/2,621)	34.4% (262/761)	30.0% (145/483)
Newly detected diabetes	10.3% (150/1,453)	17.5% (54/309)	17.9% (25/140)
Newly detected IGT	26.2% (380/1,453)	29.1% (90/309)	35.7% (50/140)
Blood pressure, mm Hg			
Systolic	128.0 ± 17.7	133.7 ± 22.0	136.4 ± 23.6
Diastolic	78.8 ± 11.0	79.6 ± 12.6	78.5 ± 12.5
Pulse pressure	49.1 ± 12.9	54.1 ± 16.6	58.0 ± 18.3
SBP/DBP ≥140/90 mm Hg	27.9% (731/2,621)	39.6% (301/761)	46.6% (225/483)

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day of a patient's interview/examination and the date of the first cardiovascular event (or the date of the last follow-up assessment for event-free patients). The assumption of proportionality of hazards was checked using log(-log(survival)) plots. A type I error probability limit of $\alpha = 0.05$ was used to indicate statistical significance. All statistical analyses were performed using SAS statistical software release version 9.4 (SAS Institute) at the Department of Public Health and Primary Care, Ghent University.

RESULTS

Of the 4,548 participants in INTERASPIRE, the 289 Egyptian patients were excluded as local analyses did not meet the requirements of laboratory standardization with the study's central laboratory. Information on eGFR was lacking in 186 patients and

on UACR in an additional 208 patients, randomly distributed from the remaining centers, leaving 3,865 patients as the final study population (Figure 1). The distribution of age and sex was comparable between those with and without complete information on eGFR and UACR. Follow-up data on the cardiovascular endpoints during a median of 1.02 years (IQR: 1.00-1.14 years) was available in 3,577 (92.5%) of these patients (Figure 1).

PATIENT CHARACTERISTICS. Clinical characteristics for the whole population and by sex are presented in Table 1. Males accounted for 79.6% and the mean age was 60.2 ± 10.2 years. Smoking was more common among males. A history of diabetes was present among 42.9% of males and 46.8% of females. Obesity was present in 23.8% of the population (men: 21.8%; women: 31.6%).

KIDNEY FUNCTION. The CKD pattern according to the KDIGO classifications is illustrated in Figure 2. A total of 67.8% of the patients were categorized as low-risk CKD whereas 19.7% had a moderate, 6.9% a high, and 5.6% a very high-risk CKD. Among the 1,244 patients with any degree of CKD, 638 (51.3%) would have been undetected if no UACR had been available. The prevalence of CKD increased by age and was higher in women than in men across all risk groups younger than the age of 75 years (Figure 3). The distribution of patient characteristics by baseline KDIGO risk category is given in Table 2. A history of diabetes and newly detected impaired glucose tolerance and diabetes were more frequent in patients with an increased CKD risk, particularly in the high-/very high-risk groups. Obesity was highest in patients at high-/very high-risk CKD but did not differ between the different CKD groups. However, rates of a high waist-to-height ratio (≥ 0.6) were substantially higher in the higher CKD-risk groups. Physical activity was inversely associated with KDIGO risk category, with sufficient levels of physical activity being reported in only 40.1% of low-risk patients and 30.0% in high- or very high-risk patients.

Drugs with cardio-renal protective characteristics were prescribed as follows: RAAS inhibitors were prescribed to 69.3% of the patients with low-, 72.5% of those with moderate-, and 70.5% of those with high-risk CKD. Of patients using RAAS inhibitors, 22.9% had a UACR ≥ 30 mg/g compared with 25.1% of those not on RAAS inhibitors ($P = 0.480$). Sodium-glucose cotransporter 2 inhibitors were prescribed to 13.5% of the patients with low-risk CKD, 20.1% of those with moderate-risk CKD, and 16.2% with high-risk CKD. A Glucagon-like peptide-1 receptor agonist had been prescribed to 1.0%, 1.6%, and 1.0% of the

patients with low-, moderate-, and high-risk CKD, respectively. Large geographic variations (Figure 4) were observed with a CKD prevalence of 52.7% in Nigeria and 45.8% in Tanzania compared with 23.8% in Poland and 23.1% in Colombia. The additional value of including UACR as well as eGFR was, with some variations, substantial in all participating countries (Figure 4).

PROGNOSTIC IMPLICATION OF CKD. The frequency of the primary event (a composite of the first of cardiovascular death [$n = 37$], hospitalization for nonfatal myocardial infarction [$n = 73$], stroke [$n = 26$], hospitalization for heart failure [$n = 59$]) during the follow-up period, stratified by age, gender, and CKD risk category, is presented in Table 3. Overall, 4.7% experienced an event with the highest incidence in the KDIGO high-risk group (men: 11.0%; women: 9.6%). The event-rate increased by age, with 20.3% of patients ≥ 75 years experiencing a primary event. Unadjusted and adjusted HRs showed that patients with high-risk CKD had a significantly higher risk of the primary event, with a multivariately adjusted HR (95% CI) of 3.46 (2.28-5.25; $P < 0.0001$) for patients at high risk. This strongly increased risk was statistically significant in both sexes (HR [95% CI]: 3.91 [2.38-6.44], $P < 0.0001$ in men; 2.25 [1.03-4.89], $P = 0.042$ in women) (Table 4). This relationship was independent of other risk factors, and consistent across age categories. A secondary endpoint, including coronary bypass surgery ($n = 17$) and percutaneous coronary interventions ($n = 117$), was also analyzed (Table 5) revealing a multivariately adjusted HR (95% CI) of 2.11 (1.49-2.99; $P < 0.0001$) for patients at high risk (Table 6).

The cumulative incidence of the primary and secondary endpoints by KDIGO category is presented in Figure 5A and 5B, showing that the risk increased consistently over time from the baseline interview (study visit). The UACR and eGFR were prognostically unfavorable independently of each other. As shown in Table 7 the prognostic information regarding the primary endpoint of the combination of eGFR and UACR (HR: 3.72 [95% CI: 2.42-5.72]) was superior to that of eGFR alone (HR: 1.79 [95% CI: 1.19-2.71]) or UACR (HR: 2.53 [95% CI: 1.59-4.03]) alone and that UACR alone was just as informative as eGFR as an isolated test. A similar pattern was evident for the secondary endpoint (Table 8).

DISCUSSION

This study, undertaken in 14 countries across all 6 WHO regions, yielded several clinically important findings: CKD is common in patients with CAD, but

TABLE 2 Continued

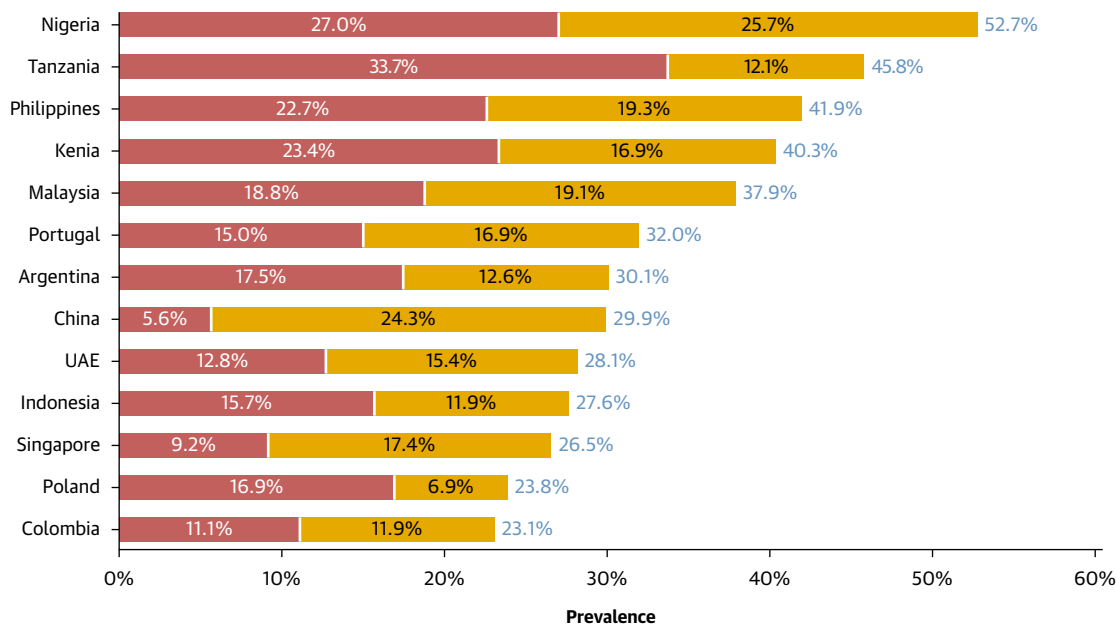
	KDIGO Risk Category		
	Low Risk (n = 2,621)	Moderately Increased Risk (n = 761)	High or Very High Risk (n = 483)
Laboratory variables			
Fasting glucose, mmol/L	6.16 $\pm$ 2.16	6.84 $\pm$ 3.04	7.46 $\pm$ 3.51
Fasting glucose ≥ 7 mmol/L	16.8% (398/2,376)	28.7% (197/686)	39.3% (161/410)
HbA1c, %	6.25 $\pm$ 1.27	6.81 $\pm$ 1.76	7.08 $\pm$ 1.97
HbA1c $\geq 5.6\%$	75.2% (1,958/2,602)	82.2% (614/747)	86.2% (407/472)
HbA1c $\geq 6.5\%$	24.9% (649/2,602)	42.2% (315/747)	50.0% (236/472)
LDL-C, mmol/L	2.22 $\pm$ 0.99	2.27 $\pm$ 1.01	2.46 $\pm$ 1.17
LDL-C ≥ 1.8 mmol/L	61.5% (1,610/2,618)	61.1% (465/761)	64.4% (311/483)
HDL-C, mmol/L	1.16 $\pm$ 0.30	1.15 $\pm$ 0.28	1.12 $\pm$ 0.30
Triglycerides, mmol/L	1.58 $\pm$ 1.01	1.68 $\pm$ 1.23	1.83 $\pm$ 1.29
Remnant cholesterol, mmol/L	0.65 $\pm$ 0.38	0.68 $\pm$ 0.42	0.75 $\pm$ 0.52
eGFR, mL/min/1.73 m ²	90.7 $\pm$ 14.5	78.7 $\pm$ 20.9	47.7 $\pm$ 24.9
Use of cardioprotective drugs			
Aspirin	94.7% (2,477/2,617)	92.2% (701/760)	89.2% (429/481)
Beta-blockers	77.2% (2,017/2,613)	77.6% (590/760)	80.7% (388/481)
RAAS blockers	64.1% (1,677/2,617)	63.4% (482/760)	64.7% (311/481)
Statins	85.3% (2,233/2,617)	83.4% (634/760)	85.4% (411/481)
SGLT-2 inhibitors	13.5% (352/2,617)	20.1% (153/760)	16.2% (78/481)
GLP-1 receptor agonists	1.0% (27/2,617)	1.6% (12/760)	1.0% (5/481)
RAAS blockers or SGLT-2 inhibitors or GLP-1 RA	69.3% (1,813/2,617)	72.5% (551/760)	70.5% (339/481)

Values are proportion (n/N) or mean $\pm$ SD.

DBP = diastolic blood pressure; eGFR = estimated glomerular filtration rate; GLP-1 = glucagon-like peptide-1; HbA1c = glycated hemoglobin; HDL-C = high-density lipoprotein cholesterol; KDIGO = Kidney Disease: Improving Global Outcomes; LDL-C = low-density lipoprotein cholesterol; RAAS = Renin Angiotensin Aldosterone System; SBP = systolic blood pressure; SGLT-2 = sodium-glucose cotransporter 2; other abbreviations as in Table 1.

more than one-half of them will be missed without UACR as part of the screening process. If present, CKD is an early-onsetting, independent risk for a poorer prognosis in CAD patients (Central Illustration). Yet the use of cardio-renal protective treatment is suboptimal.

The main objective of INTERASPIRE was to measure in patients with CAD, identified from countries across all 6 WHO regions, if guideline standards for CVD prevention are being achieved. This study on CKD extends our report from a European population¹⁵ to countries from the other 5e WHO regions (including a sample of patients from middle-income countries) and demonstrates the high prevalence of CKD in CAD patients around the world. Further, it underlines the importance of combining eGFR with UACR in the screening process of all CAD patients. In both studies about one-half of the patients with CKD would have remained undetected without information on the UACR.

FIGURE 4 Prevalence of CKD by Participating Country (Blue Percentages) and the Additional Value of Including UACR Beyond eGFR

Red bars = CKD based on eGFR <60 mL/min/1.73 m²; orange bars = CKD based on UACR ≥30 mg/g alone. UAE = United Arab Emirates; other abbreviations as in Figures 2 and 3.

Albuminuria represents an early sign of endothelial dysfunction, ie, vascular involvement,^{20,21} whereas eGFR may be seen as an expression of a more advanced, structural kidney engagement. Thus, these 2 expressions of kidney involvement complement each other.

As expected, the presence of CKD increased by age, and was more common in women. Importantly, progression to kidney failure has been reported to be more common in men, who also are at a higher cardiovascular risk than women.²² A factor associated with CKD was, not unexpectedly,¹³ the presence of diabetes. Body mass index was higher in high-/very high-risk CKD but may be less valid in a sample including patients of Asian origin. Waist-to-height ratio, which differed substantially between the CKD groups, may then be a more appropriate expression for abdominal overweight.²³ Insufficient physical activity was more frequent in participants with moderate- or high-risk CKD. Similar observations were made in a European cohort.¹⁵

The overall prevalence of CKD in the present sample was 32.3% with substantial geographic variations. Our data do not provide a clear explanation for these differences in prevalence between countries. One potential explanation is the more frequent representation and less efficient management of patients with diabetes in countries with a higher proportion of CKD.²⁴ The reasons may, however, also reflect true regional and/or ethnic differences. In Southeast Asia there is a larger burden of cardio-metabolic disease which may be related to genetic

TABLE 3 Cumulative Incidence of the Primary Endpoint, a Composite of the First of Cardiovascular Death, Hospitalization for Nonfatal Myocardial Infarction, Stroke, or Heart Failure by Age, Gender, and Risk Category

	All (N = 3,570)	KDIGO Category		
		Low Risk (n = 2,434)	Moderately Increased Risk (n = 700)	High Risk (n = 436)
All	4.7% (167/3,570)	3.2% (79/2,434)	6.0% (42/700)	10.6% (46/436)
Gender				
Men	4.3% (122/2,849)	2.8% (56/2,006)	6.1% (33/543)	11.0% (33/300)
Women	6.2% (45/721)	5.4% (23/428)	5.7% (9/157)	9.6% (13/136)
Age				
<55 y	4.1% (44/1,069)	3.4% (28/823)	5.1% (9/176)	10.0% (7/70)
55-64 y	3.8% (48/1,258)	2.7% (24/886)	6.5% (14/215)	6.4% (10/157)
65-74 y	5.1% (51/998)	3.7% (23/625)	5.6% (13/233)	10.7% (15/140)
≥75 y	9.8% (24/245)	4.0% (4/100)	7.9% (6/76)	20.3% (14/69)

Values are proportion (n/N).

TABLE 4 Unadjusted and Adjusted HRs for Primary Outcomes, a Composite of the First of Cardiovascular Death, Hospitalization for Nonfatal Myocardial Infarction, Stroke, or Heart Failure, By Risk Category

KDIGO Risk Category	HR (95% CI), P Value		
	Unadjusted	Adjusted for Age and Gender	Multivariately Adjusted ^a
All			
Low	1	1	1
Moderately increased	1.89 (1.30-2.76), $P < 0.0001$	1.89 (1.30-2.76), $P < 0.0001$	2.10 (1.40-3.14), $P < 0.0001$
High	3.25 (2.23-4.73), $P < 0.0001$	3.25 (2.23-4.73), $P < 0.0001$	3.46 (2.28-5.25), $P < 0.0001$
Men			
Low	1	1	1
Moderately increased	2.26 (1.47-3.48), $P < 0.0001$	2.21 (1.43-3.41), $P < 0.0001$	2.36 (1.48-3.78), $P < 0.0001$
High	4.17 (2.71-6.41), $P < 0.0001$	4.02 (2.58-6.25), $P < 0.0001$	3.91 (2.38-6.44), $P < 0.0001$
Women			
Low	1	1	1
Moderately increased	1.28 (0.59-2.79), $P = 0.528$	1.27 (0.58-2.76), $P = 0.548$	1.37 (0.61-3.10), $P = 0.445$
High	2.16 (1.09-4.29), $P = 0.028$	2.09 (1.04-4.20), $P = 0.038$	2.25 (1.03-4.89), $P = 0.042$

predisposition in combination with differences in diet, lifestyle, and socioeconomic status.²⁵ Individuals of African ancestry are disproportionately affected by kidney disease, which is related to the Apolipoprotein L1 (*APOL1*) gene.^{26,27} Further studies are needed to explain these substantial variations in the prevalence of CKD. What is consistent across all countries is the value of adding UACR to eGFR to detect the true prevalence of CKD as outlined in **Figure 4**.

The risk associated with under-recognized CKD emerged early, apparent already after a rather short follow-up demonstrating early risk emergence rather than long-term progression. The poorer prognosis of CKD was independent of other risk factors, highlighting the importance of screening soon after the diagnosis of CAD to ensure prescription of cardio-renal protective therapy. In the Dutch PREVEND (Prevention of Renal and Vascular Endstage Disease) study,²⁸ the UACR was the strongest predictor in identifying individuals at risk of cardiovascular morbidity and mortality. In the present study eGFR and UACR were independent predictors of a poor outcome, but the combination provided the strongest prognostic value. Currently, international guidelines for the prevention and management of patients with CAD do not recommend screening with both UACR and eGFR.^{14,29} CKD is mostly asymptomatic, which will contribute to its under-recognition in CAD, together with the limited guidance for screening of this patient population.¹¹ Failure to measure albuminuria represents a missed opportunity to recognize residual cardio-renal risk. As the practice of cardiology increasingly integrates kidney-protective therapies, UACR measurement deserves consideration as part of routine CAD care. Our findings are in favor of

introducing a recommendation in CAD management guidelines to systematically screen for CKD by means of a combination of eGFR and UACR.

Pharmacologic interventions can retard the progression of CKD and reduce cardiovascular events. The present data on the use of drugs with cardio-renal protective capacity cannot distinguish between the use in previously undiagnosed CKD and CKD that was known but not documented. Moreover, the indication to prescribe may have been for cardiovascular reasons. RAAS inhibitors, prescribed to almost two-thirds of the present sample of CAD patients, have been reported to reduce the risk of cardiovascular events.³⁰ Thus, it was somewhat surprising that the presence of a UACR ≥ 30 mg/g did not differ between patients prescribed such therapy

TABLE 5 Cumulative Incidence of the Secondary Endpoint, a Composite of the First of Cardiovascular Death, Hospitalization for Nonfatal Myocardial Infarction, Stroke, Heart Failure, Coronary Bypass Surgery, or Percutaneous Coronary Intervention by Age, Gender, and Risk Category

	KDIGO Category			
	All (N = 3,577)	Low Risk (n = 2,438)	Moderately Increased Risk (n = 703)	High Risk (n = 436)
All	7.9% (282/3,577)	6.4% (157/2,438)	10.0% (70/703)	12.6% (55/436)
Gender				
Men	7.7% (219/2,852)	6.0% (121/2,007)	10.8% (59/545)	13.0% (39/300)
Women	8.7% (63/725)	8.4% (36/431)	7.0% (11/158)	11.8% (16/136)
Age				
<55 y	7.2% (77/1,066)	6.8% (56/820)	7.4% (13/176)	11.4% (8/70)
55-64 y	7.0% (89/1,266)	5.4% (48/895)	12.6% (27/215)	9.0% (14/156)
65-74 y	8.5% (85/998)	7.7% (48/622)	8.9% (21/235)	11.3% (16/141)
≥ 75 y	12.6% (31/247)	5.0% (5/101)	11.7% (9/77)	24.6% (17/69)

Values are proportion (n/N).
Abbreviations as in **Table 2**.

TABLE 6 Unadjusted and Adjusted HRs for the Secondary Outcome, a Composite of the First of Cardiovascular Death, Hospitalization for Nonfatal Myocardial Infarction, Stroke, Heart Failure, Coronary Bypass Surgery, or Percutaneous Coronary Intervention, by Risk Category

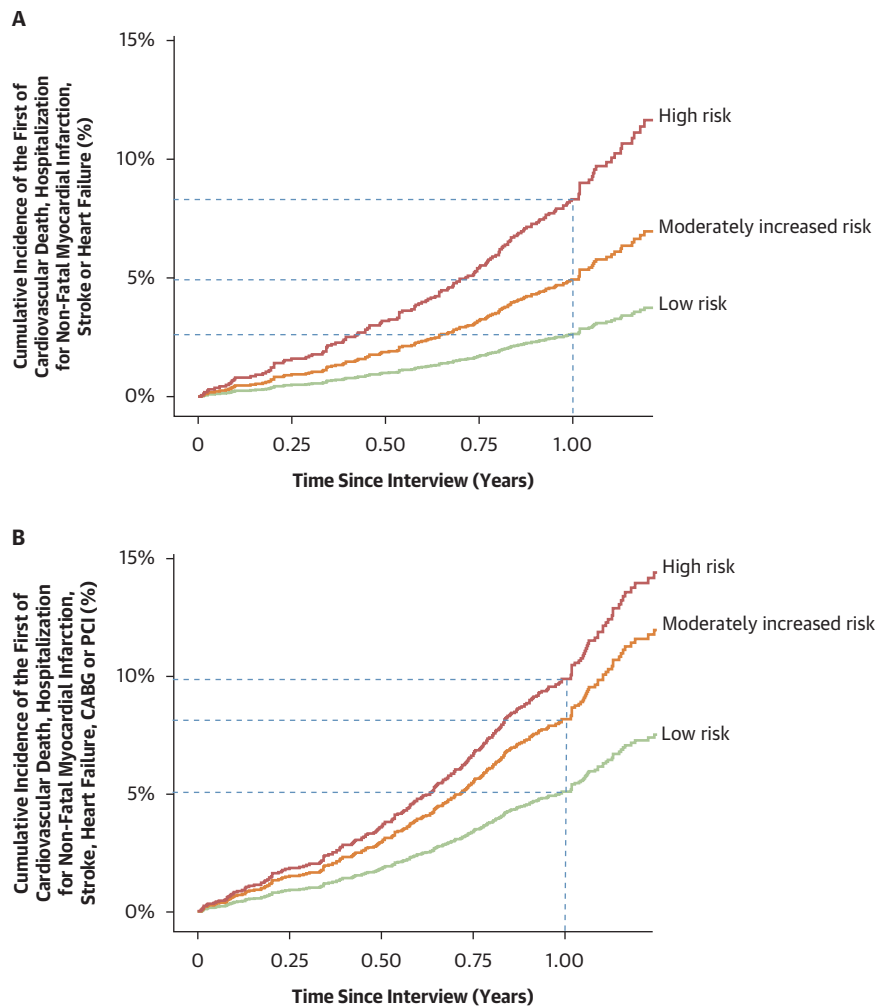
KDIGO Risk Category	HR (95% CI), P Value		
	Unadjusted	Adjusted for Age and Gender	Multivariately Adjusted ^a
All			
Low	1	1	1
Moderately increased	1.67 (1.26-2.21), <i>P</i> < 0.0001	1.63 (1.23-2.17), <i>P</i> < 0.0001	1.78 (1.31-2.41), <i>P</i> < 0.0001
High	2.07 (1.52-2.82), <i>P</i> < 0.0001	1.99 (1.45-2.72), <i>P</i> < 0.0001	2.11 (1.49-2.99), <i>P</i> < 0.0001
Men			
Low	1	1	1
Moderately increased	1.91 (1.40-2.61), <i>P</i> < 0.0001	1.88 (1.37-2.57), <i>P</i> < 0.0001	2.03 (1.44-2.85), <i>P</i> < 0.0001
High	2.25 (1.56-3.22), <i>P</i> < 0.0001	2.18 (1.50-3.15), <i>P</i> < 0.0001	2.16 (1.43-3.25), <i>P</i> < 0.0001
Women			
Low	1	1	1
Moderately increased	0.97 (0.49-1.92), <i>P</i> = 0.934	0.96 (0.48-1.89), <i>P</i> = 0.897	1.02 (0.50-2.06), <i>P</i> = 0.963
High	1.63 (0.90-2.95), <i>P</i> = 0.105	1.56 (0.86-2.85), <i>P</i> = 0.147	1.59 (0.81-3.13), <i>P</i> = 0.178

^aAdjusted for age at interview, gender, history of myocardial infarction, history of percutaneous coronary intervention, history of coronary bypass surgery, history of stroke, history of diabetes, smoking, waist-to-height ratio, physical activity, systolic blood pressure, fasting glucose, HbA1c, LDL cholesterol, use of aspirin, use of beta-blockers, use of RAAS blockers, and use of statins.
Abbreviations as in Table 2.

or not, and that a statistical adjustment for the use did not decrease the negative prognostic effect of CKD. More recently introduced drugs with cardio-renal protective capacity are sodium-glucose cotransporter 2 inhibitors, glucagon-like peptide-1 receptor agonists and Finerenone.³¹⁻³⁵ Sodium glucose cotransporter 2 inhibitors were prescribed to 1 of 5 patients with moderate- or high-risk CKD, whereas the use of the other 2 agents was negligible. From a clinical point of view this opens an opportunity to improve patient management. In this context, it should be emphasized that the INTERASPIRE study does not prove that screening for CKD improves outcomes. Rather, it underlines the importance of screening for CKD in patients with CAD with the combined use of UACR and eGFR to identify patients at greater risk, who are eligible for cardio-renal therapies with proven benefit.

STUDY STRENGTHS AND LIMITATIONS. A key strength of INTERASPIRE is the rigorous data collection, conducted by trained staff using standardized interviews, measurement tools, and protocols. Another strength is the inclusion of countries from all WHO regions outside Europe, resulting in an international perspective of kidney function in CAD patients. It may be seen as if women are under-represented. They do, however, develop manifestations of CAD later than men, which makes the populations representative for an unselected

population of such patients aged <80 years. This survey was limited to selected geographic areas and hospitals that may not be nationally representative and to the extent that centers were specialized in cardiology. The data may, therefore, underestimate the actual screening for CKD in CAD patients. Moreover, a survival bias cannot be excluded because we studied survivors after 6 months to 2 years after their hospital admission. Another limitation may be that UACR was estimated locally. The tests were, however, performed according to routine standards in each hospital. Only 1 examination of the UACR was available. The present results underline the importance of UACR examinations for valid screening of CKD, but repeated samples are needed to verify the findings in clinical practice. Nevertheless, a single UACR was sufficient to identify patients who warrant further evaluation and intervention. The study, planned in 2019 and launched during the COVID-19 pandemic, faced a challenge given the requirement for face-to-face interviews and examinations. This may have caused selection bias but conservative as sicker patients may have been less willing to attend hospital. Thus, a truly representative picture of patient management is likely to be even poorer. Moreover, the results are consistent with European data achieved before the pandemic.¹⁵ Despite these limitations, our study highlights important aspects on the screening and management of CAD patients for CKD worldwide.

FIGURE 5 Cumulative Incidence

(A) Cumulative incidence of the primary endpoint, a composite of the first of cardiovascular death, hospitalization for nonfatal myocardial infarction, stroke, or heart failure by KDIGO risk category. (B) Cumulative incidence of the secondary endpoint, a composite of the first of cardiovascular death, hospitalization for nonfatal myocardial infarction, stroke, heart failure, coronary bypass surgery, or percutaneous coronary intervention by KDIGO risk category. CABG = coronary artery bypass graft; PCI = percutaneous coronary intervention; other abbreviations as in [Figure 2](#).

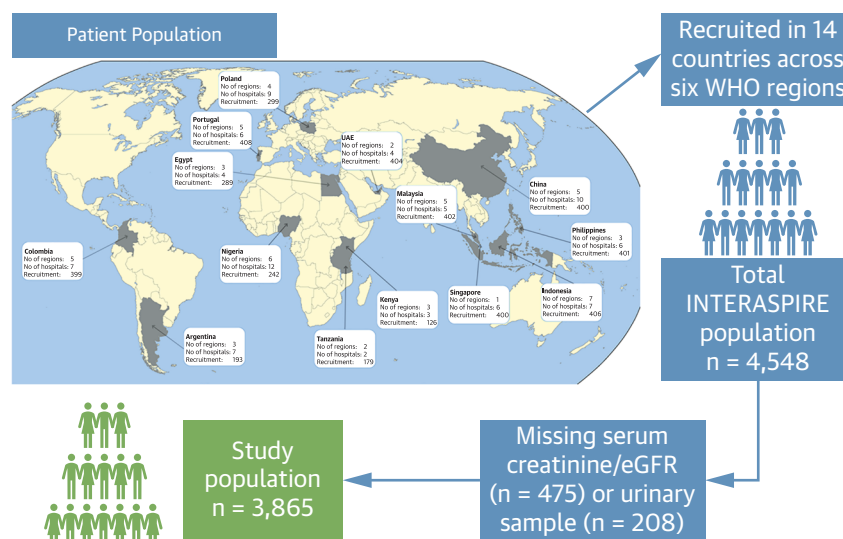
TABLE 7 Prognostic Information of the Primary Outcome, a Composite of the First of Cardiovascular Death, Hospitalization for Nonfatal Myocardial Infarction, Stroke, or Heart Failure, by eGFR and UACR as Isolated and Combined Entities

Primary Outcome	HR, Adjusted for Age and Gender (95% CI)
eGFR ≥ 60 mL/min/1.73 m ² and UACR < 30 mg/g	1
eGFR < 60 mL/min/1.73 m ² and UACR < 30 mg/g	1.79 (1.19-2.71), $P < 0.0001$
eGFR ≥ 60 mL/min/1.73 m ² and UACR ≥ 30 mg/g	2.53 (1.59-4.03), $P = 0.0055$
eGFR < 60 mL/min/1.73 m ² and UACR ≥ 30 mg/g	3.72 (2.42-5.72), $P < 0.0001$

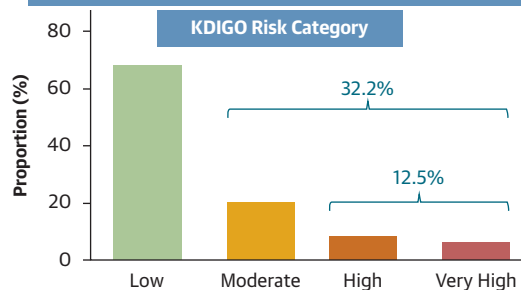
TABLE 8 Prognostic Information of the Secondary Outcome, a Composite of the First of Cardiovascular Death, Hospitalization for Nonfatal Myocardial Infarction, Stroke, Heart Failure, Coronary Bypass Surgery, or Percutaneous Coronary Intervention, by eGFR and UACR as Isolated and Combined Entities

Secondary Outcome	HR, Adjusted for Age and Gender (95% CI), P Value
eGFR ≥ 60 mL/min/1.73 m ² and UACR < 30 mg/g	1
eGFR < 60 mL/min/1.73 m ² and UACR < 30 mg/g	1.32 (0.95-1.84), $P = 0.100$
eGFR ≥ 60 mL/min/1.73 m ² and UACR ≥ 30 mg/g	2.24 (1.58-3.19), $P < 0.0001$
eGFR < 60 mL/min/1.73 m ² and UACR ≥ 30 mg/g	2.28 (1.58-3.30), $P < 0.0001$

eGFR = estimated Glomerular Fraction; UACR = Urinary Albumin Excretion Ratio.

CENTRAL ILLUSTRATION The INTERASPIRE Study of CKD and Coronary Heart Disease

Chronic Kidney Disease (CKD) Assessed According to Kidney Disease Improving Global Outcomes (KDIGO) by Combining eGFR With Urinary Albumin Creatinine Ratio (UACR)



51.3% of those with CKD had been undetected without UACR

**Prognostic implication of CKD**

Primary endpoint: the first of cardiovascular death, hospitalization for non-fatal myocardial infarction, stroke or heart failure

eGFR <60 mL/min/1.73 m² and UACR <30 mg/g = HR 1.79 (1.19-2.71), P = 0.0055
 eGFR ≥60 mL/min/1.73 m² and UACR ≥30 mg/g = HR 2.53 (1.59-4.03), P = 0.0055
 eGFR <60 mL/min/1.73 m² and UACR ≥30 mg/g = HR 3.72 (2.42-5.75), P ≤ 0.0001

Al-Azzawy SM, et al. JACC. 2026;■(■):■-■.

CKD = chronic kidney disease; eGFR = estimated glomerular filtration rate; UACR = urinary albumin creatinine ratio; WHO = World Health Organization.

CONCLUSIONS

Early screening for CKD in patients with CAD is important and must include both eGFR and UACR to provide adequate information. Without UACR, one-half of the patients with CKD would remain undetected. Treatment with cardio-renal protective therapy was low, indicating a significant improvement potential.

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